Financial Evaluation of the AEB Rabbit Testing-as-a-Service Business

Harry Emes

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Executive Summary

The AEB Rabbit is a supercapacitor-powered Automated Emergency Braking (AEB) target vehicle engineered for explosive, six-second "panic" manoeuvres on behalf of autonomous-vehicle (AV) developers. It is commercialised as a Testing-as-a-Service (TaaS) / Product–Service System [1]: the operator retains ownership of the fleet and sells test-days to AV developers, who absorb none of the capital or maintenance burden. The market analysis [2] estimates a $1.5–2 bn global AV-validation market; this chapter evaluates whether the specific financial structure of the TaaS build-out – front-loaded CAPEX, utilisation-tethered revenue, maintenance-dependent uptime, and terminal-value-dominated returns – clears standard capital-budgeting thresholds.[1]

Headline financial evaluation:

- **CAPEX commitment:** £0.98M over five years (£310k per vehicle $\times$ 3 vehicles + £50 k Y1 setup). Cumulative revenue first exceeds cumulative CAPEX in Y2.
- **Five-year trajectory:** £300k revenue $\rightarrow$ £2.1M; break-even in Y4 at 122 test days/year (contribution £5,650 per day).
- **Funding:** £1.0M seed + £1.5M Series A, total £2.5M equity. Peak cumulative cash outflow occurs end-Y2 ($\sim$£90 k residual).
- **Capital return:** project NPV £1.32M at a 20% VC discount rate, £0.97M at a 25% hurdle rate, IRR 55.2%, all including a £5.2M terminal value at $2.5\times$ Y5 revenue.
- **Sensitivity:** day-rate and utilisation dominate; a $\pm 20\%$ swing in either drives $\sim$£270–370 k in Y3 EBIT.
- **Downtime sensitivity:** NPV stays positive through $> 30\%$ unplanned fleet downtime, but Y5 EBIT turns negative once downtime exceeds $\sim$20%. Fleet-maintenance protocols are therefore the single most binding execution dependency on the financial plan.

Conclusion: the TaaS financial structure clears standard venture-capital thresholds in the base case and is robust to Bull/Bear excursions. The business is most exposed to compounded shortfalls in day-rate and utilisation (external) or fleet downtime (internal), and financial viability in both directions is dominated by a single assumption – the peer-group terminal-value multiple. Recom-

[1] This chapter, the Python financial model, all figures and prose, is the sole work of Harry Emes. Market material cited here is the work of Toby McConnell [2]; prototype and supply-chain context is owed to the engineering chapters of the group report.

mendations in Chapter 6.

Chapter 1: Scope and methodology

1.1 Why the TaaS structure requires dedicated financial evaluation

Conventional automotive-testing equipment is sold as a capital good: the OEM or testbed buyer pays once, depreciates the asset, and owns both the utilisation upside and the maintenance downside. The AEB Rabbit is instead delivered as a Testing-as-a-Service / Product Service System [1, 3]: the operator retains the asset, sells the usage, and absorbs the full capital and maintenance risk. This inverts the financial structure of the business in three ways that conventional hardware-sale financial analysis does not capture:

1. **CAPEX inversion.** The full fleet must be built and made operational *before* a billable hour is executed. The operator therefore finances the asset across its useful life from revenue that only flows if utilisation hits plan.
2. **Utilisation-tethered revenue.** Every day the fleet is not earning revenue is a day the fleet is still depreciating, consuming fixed cost, and eroding unit economics. Revenue is a function not just of customer demand but of operational availability.
3. **Terminal-value dominance.** The five-year operational free cash flow is loss-making until Y4; the NPV is almost entirely the present value of the Y5 exit (terminal value). Conventional accounting-profit analysis is insufficient to assess investability.

This chapter therefore extends the standard financial-plan toolkit (cost model, P&L, cash flow) with three TaaS-specific analyses: a CAPEX-inversion timeline (Section 2.4), a utilisation-tethered break-even + sensitivity (Sections 3.1–3.3), and a full NPV / IRR capital-budgeting assessment (Section 3.2) with a dedicated downtime-sensitivity treatment (Section 4.1) to quantify the cost of maintenance-protocol failure.

1.2 Model scope and handoffs

The financial model is the single source of truth for every number in this chapter: a driver-based Python script emits tables, figures and in-line macros from one shared driver set, so a change to Y1 utilisation regenerates the whole plan consistently. The methodology mirrors the engineering chapter's simulation toolchain [4], where a single simulator drives the sensitivity ranking, compliance arbiter, and final-vehicle prediction.

The chapter is deliberately scoped to financial evaluation and depends on four external inputs from co-authors, cited inline throughout rather than re-derived:

- **Market sizing, competitor day-rate benchmarks, segmentation and phased customer targeting** [2].
- **Prototype functional requirements, PAS-standards compliance timeline, and ALARP-safety validation cycles** [5, 6, 7, 8] (Ziyang Xing, engineering chapter).
- **Fleet supply-chain design, $<$30-minute quick-recovery protocol, and 9R-framework refurbishment loops** (Yuze Jiang, supply-chain chapter).
- **Fleet maintenance protocols and downtime assumptions** (David, reliability chapter).

The financial analysis is then validated against the market-access gatekeepers identified by McConnell: CAM Testbed UK, the Centre for Connected and Autonomous Vehicles (CCAV), and Thatcham Research [9, 10, 11] – their acceptance is a hard gate on OEM-scale revenue and drives the Y3-onwards revenue trajectory.

1.3 Document roadmap

Chapter 2 specifies the CAPEX-inverted cost model, the utilisation-driven revenue model, and the resulting five-year P&L and cash-flow forecast. Chapter 3 applies capital-budgeting analysis: break-even, NPV, IRR, and scenario analysis. Chapter 4 treats the financial risk lens, including the downtime-sensitivity analysis Paul's brief calls out explicitly. Chapter 5 closes with the funding strategy, valuation discussion, and the financial KPIs the investor should track.

Chapter 2: Cost structure, revenue model and five-year forecast

2.1 CAPEX model: the cost of ownership the operator absorbs

The operator owns the fleet; the customer rents days. Every pound of AEB Rabbit capital therefore sits on our balance sheet, is financed from equity until the business turns cash-positive, and is recovered through the day-rate. Per-vehicle build is £310k, itemised in Table 2.1; the Y1 setup adds £50 k in workshop fit-out and data-platform infrastructure. Fleet growth (2nd vehicle in Y3, 3rd in Y5) drives a further £620 k of capex over Y3–Y5, summarised in Table 2.2. Total capex over the five-year plan: £0.98M.

Table 2.1: Per-vehicle capex build-up and Y1 one-off setup. Figures are internal team estimates benchmarked against published Formula Student cost reports and published industrial-test-rig build quotes.

Line item	Cost (£k)
Vehicle build (chassis, powertrain, aero)	180
Data acquisition & instrumentation	60
Safety & emergency redundancy systems	25
Trailer & transport rig	30
Spares, tooling & initial tyre stock	15
Total per vehicle	**310**
Workshop fit-out & office setup (Y1 only)	35
Data platform & analytics stack (Y1 only)	15

Table 2.2: Five-year capex schedule (£k). Per-vehicle build costs £310k (see Table 2.1); one-off Y1 company setup £50k.

Year	New vehicles	Vehicle capex	Other capex	Total
1	1	310	50	**360**
2	0	0	0	**0**
3	1	310	0	**310**
4	0	0	0	**0**
5	1	310	0	**310**

Capital is depreciated on a five-year straight-line basis. No residual value is modelled at Y5: in a PSS business, the physical asset is either refurbished in place under Yuze Jiang's 9R-framework loop (and therefore stays in service beyond the modelled window) or retired in line with safety-critical-testing industry norms.

2.2 OPEX model: a fixed-cost-heavy delivery operation

Fixed opex is dominated by headcount (60% of the Y1 opex base of £538k), with the remainder split across workshop rent, insurance, certification fees, data-platform hosting, marketing, vehicle maintenance, and professional services. Table 2.3 shows the Y1 build-up; the model inflates fixed opex at 3% per year (UK CPIH 2023–25 trend) and adds step-changes for each fleet-growth hire.

Table 2.3: Year-1 fixed opex build-up (£k). Salaries benchmarked against 2026 Gradcracker/Reed automotive-engineering bands for UK regional centres; workshop rent from 2024 average Oxford-Didcot light-industrial rates.

Line item	Cost (£k)
CEO / Commercial Director (1 FTE)	75
Operations Manager / Lead Engineer (1 FTE)	60
Test Engineer (2 FTE @ £50k)	100
Data & Software Engineer (1 FTE)	60
Admin & finance (0.3 FTE)	15
Employer NI & benefits (20% of salary)	62
Workshop & office rent (UK regional)	36
Insurance (public, product, vehicle)	40
Certification & standards fees	10
Marketing & business development	30
Software, IT & data platform hosting	15
Vehicle maintenance contract	20
Professional services (legal, audit, R&D)	15
Total fixed opex, Y1	**538**

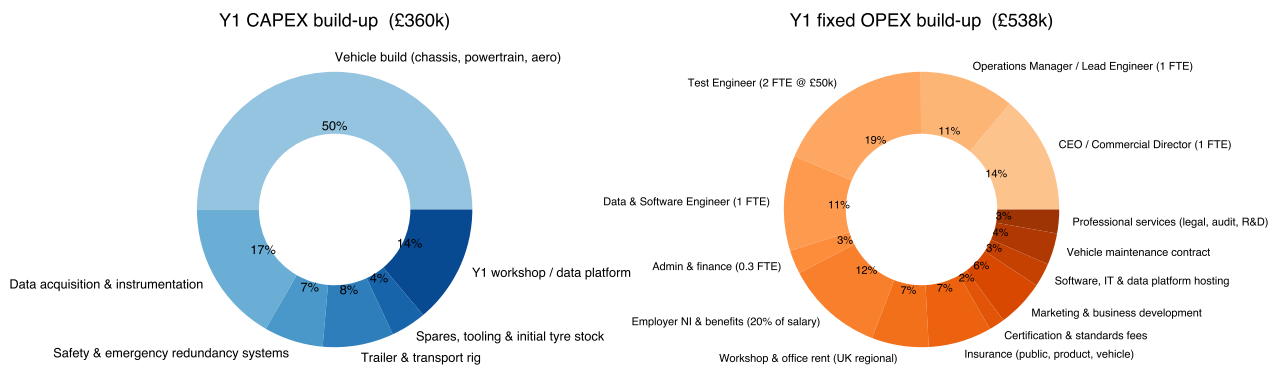

Figure 2.1: Y1 CAPEX build-up (left) and Y1 fixed-OPEX build-up (right). The opex donut shows that 60% of Y1 fixed costs are salaries plus employer NI – the reason utilisation (revenue per engineer per year) is the highest-leverage financial driver in the sensitivity analysis (Section 3.3).

Variable cost per test day is £2,300 (Y1), comprising tyres and consumables (£800), transport (£600), crew travel-and-subsistence (£400), and amortised vehicle wear (£500). The resulting contribution margin per test day is £5,650 at Y2 – this is the single number that drives the break-even analysis below.

2.3 Revenue model: every pound tethered to fleet utilisation

Revenue is modelled bottom-up from three driver arrays: test days sold, fleet size, and blended day-rate. Day-rate is the mix-weighted average of the Bronze / Silver / Gold packages (£6 k / £8 k / £10 k) benchmarked against the competitor set in McConnell's analysis [2] (AB Dynamics [12], UTAC [13], HORIBA [14], and AVL [15]). Mix-shift from Bronze-heavy to Gold-heavy over five years drives blended day-rate from £7.5 k (Y1) to £9.5 k (Y5) without any individual-tier price movement – important for the pricing discipline discussed in Section 3.3.

Table 2.4: Revenue build-up: test days, fleet utilisation, blended day rate, and resulting gross-profit stack.

Metric	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**
Test days sold	40	70	110	150	220
Fleet size	1	1	2	2	3
Utilisation (days/vehicle)	40	70	55	75	73
Blended day rate (£)	7,500	8,000	8,500	9,000	9,500
Revenue (£k)	300	560	935	1,350	2,090
Variable cost (£k)	92	164	264	368	550
Gross profit (£k)	208	396	671	982	1,540
Gross margin (%)	69.3	70.6	71.8	72.8	73.7

Reconciliation with McConnell's SOM. Y5 forecast revenue of £2.1M (≈$2.7 M) sits at 10–14% of McConnell's serviceable-obtainable-market estimate of $26 M [2] – a penetration consistent with a credible early-stage entrant and providing headroom for the bull-scenario analysis in Section 3.4.

2.4 CAPEX inversion timeline

Figure 2.2 makes the TaaS CAPEX-inversion concrete. Annual CAPEX is heavily front-loaded (Y1 = £360 k) and then steps at each fleet expansion; annual revenue ramps gradually with utilisation. Cumulative revenue crosses cumulative CAPEX in Y2– faster than a naive reading of the plan would suggest, because the pilot-year utilisation of 40 test days already generates £300k against a Y1 CAPEX spend of £360 k.

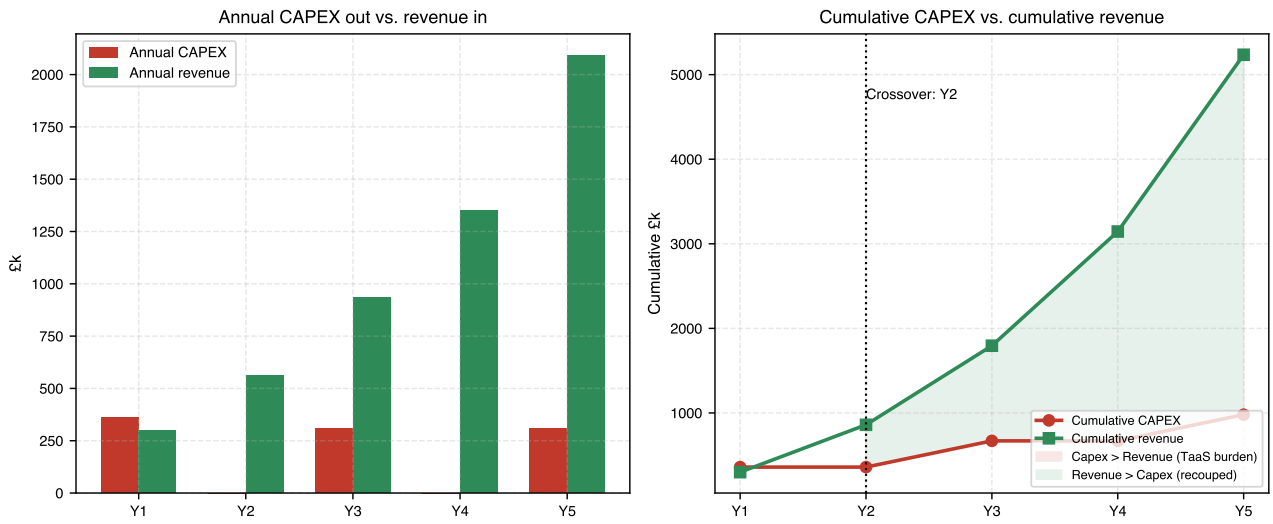

Figure 2.2: CAPEX-inversion timeline. Left: annual CAPEX out vs. annual revenue in. Right: cumulative CAPEX (red) vs. cumulative revenue (green). The red-shaded region is the "TaaS burden" period during which cumulative CAPEX exceeds cumulative revenue; the green region is the "recouped" period after Y2.

The inversion analysis should not be confused with profitability: cumulative revenue crosses cumulative CAPEX in Y2 but cumulative *gross profit* does not cross cumulative CAPEX until Y3, and cumulative *EBIT* does not cross cumulative CAPEX until Y5. The CAPEX inversion is therefore a necessary but not sufficient condition for financial viability; the break-even and NPV analyses in Chapter 3 assess the sufficient condition.

2.5 Five-year profit & loss forecast

Table 2.5 is the five-year P&L summary; every column is a downstream result of the cost and revenue models above. Three features are worth noting:

- **EBITDA turns positive in Y4**, well before net profit, reflecting the drag of depreciation in a capex-heavy service business.
- **Operating-leverage tipping point at Y3.** Gross profit grows faster than fixed opex from Y3 onwards; Y3 is the *commercial* break-even year, Y4 is the *accounting* break-even year.
- **Corporation tax kicks in at Y4** (UK small-profits rate 19% in the 2025/26 regime). The Y5 net profit of £293 k is reported after tax.

Table 2.5: Five-year P&L summary (£k). All figures flow from the single driver set stated in Sections 2.1–2.3.

Line	Y1	Y2	Y3	Y4	Y5
Revenue	300	560	935	1,350	2,090
Variable cost	-92	-164	-264	-368	-550
Gross profit	208	396	671	982	1,540
Fixed opex	-538	-616	-693	-834	-983
EBITDA	-330	-220	-22	149	557
Depreciation	-72	-72	-134	-134	-196
EBIT	-402	-292	-156	15	361
Tax @ 19%	-0	-0	-0	-3	-69
Net profit	-402	-292	-156	12	293

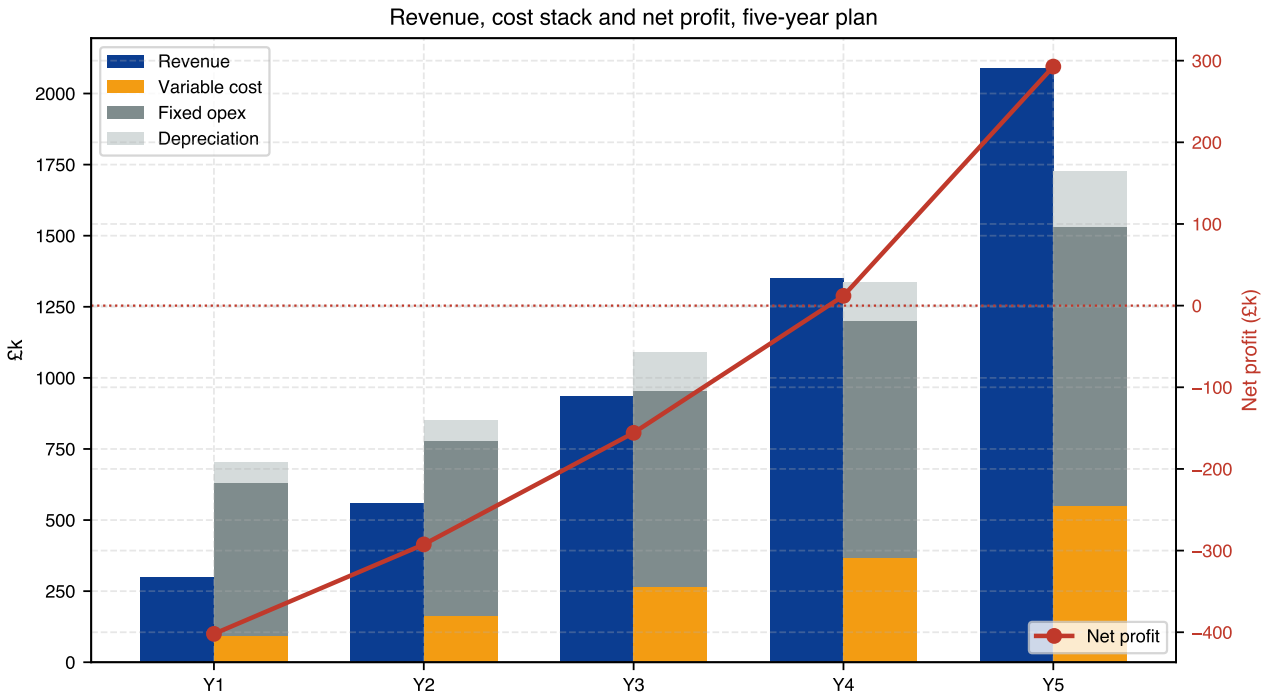

Figure 2.3: Five-year revenue (blue) against the full cost stack (variable, fixed, depreciation) with net-profit trajectory overlaid (red, right axis). Net profit crosses zero in Y4 – the accounting break-even year – and reaches £293 k net of tax in Y5.

2.6 Five-year cash-flow forecast

A capex-heavy TaaS business lives or dies on cash, not on accounting profit. Table 2.6 tracks free cash flow, equity raises, and cumulative cash balance.

Table 2.6: Five-year cash-flow forecast (£k). Seed round £1.0M at Y1; Series A £1.5M at Y3.

Line	Y1	Y2	Y3	Y4	Y5
Net profit	-402	-292	-156	12	293
+ Depreciation	72	72	134	134	196
Operating cash flow	-330	-220	-22	146	489
- Capex	-360	-0	-310	-0	-310
Free cash flow	-690	-220	-332	146	179
+ Equity raised	1,000	0	1,500	0	0
Net cash flow	310	-220	1,168	146	179
Cumulative cash	310	90	1,258	1,404	1,582

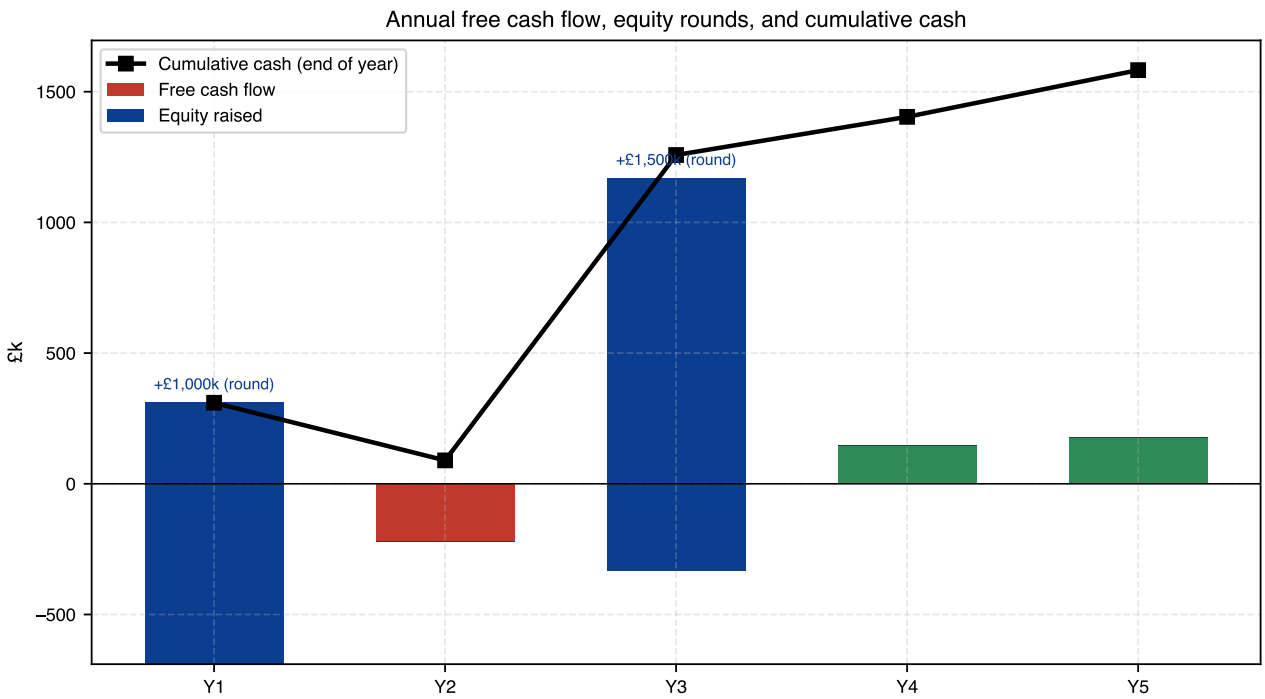

Figure 2.4: Annual free cash flow (green/red bars), equity rounds (blue stacked bars), and cumulative cash balance (black line). Cumulative cash reaches its minimum of £90 k at end-Y2 – the trigger point for Series A close – and turns structurally positive from Y3 onwards once the Series A has closed.

Two moments deserve attention. First, end-Y2 cash balance is £90 k – the tightest point in the plan, after absorbing two years of operating loss. A delayed Series A would push the company into a bridge-round conversation; the Series A therefore targets a Q4-Y2 close rather than a Y3 close. Second, Y5 free cash flow is £179 k positive even after the £310k third-vehicle purchase – the plan

is genuinely self-funding by Y5.

Chapter 3: Capital-budgeting assessment

3.1 Break-even analysis

Break-even on the Y2 fixed-cost base occurs at 122 test days per year. This is the number of days the fleet must sell annually to cover fixed opex and depreciation:

$$\mathrm{BE}_{\mathrm{days}} = \frac{F + D}{p_{\mathrm{day}} - c_{\mathrm{var}}} \tag{3.1.1}$$

where F is annual fixed opex, D annual depreciation, p_{day} the Y2 blended day-rate, and c_{var} the Y2 variable cost per day. The resulting contribution margin per day is £5,650; fixed costs plus depreciation total £690 k Y2.

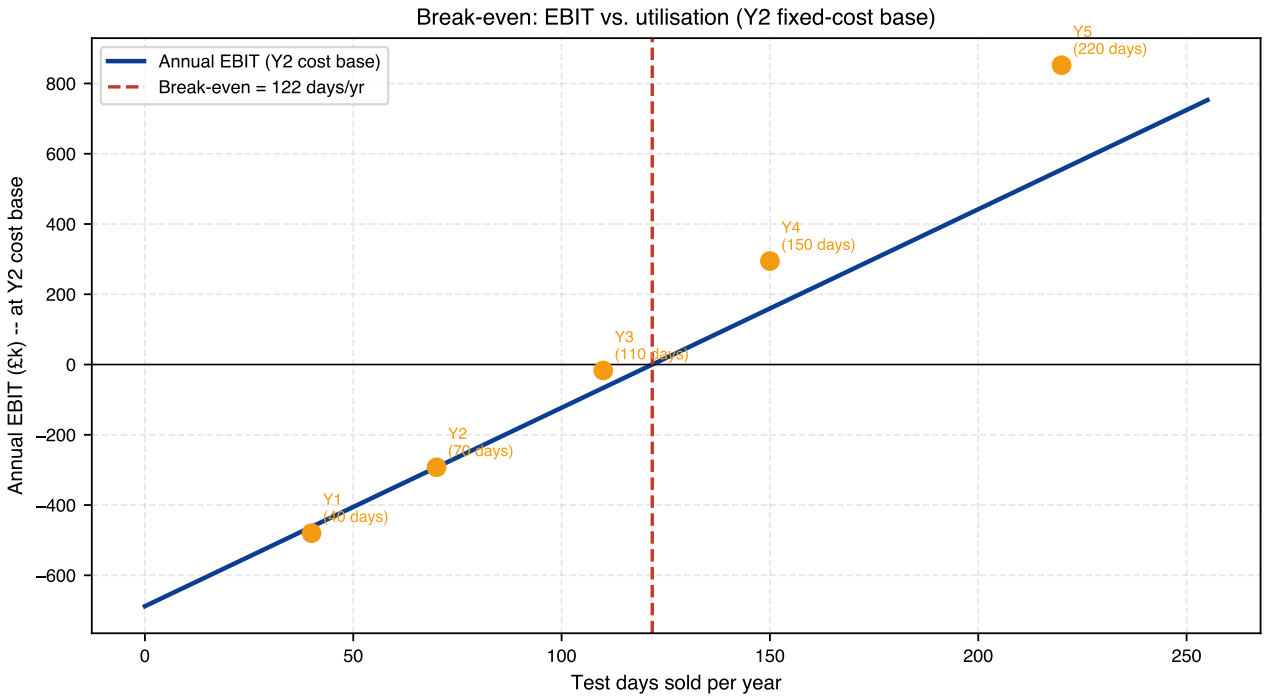

Figure 3.1: Break-even curve – annual EBIT at the Y2 cost base as a function of annual test days. The vertical red dashed line marks the 122-day break-even point; orange markers show the plan's Y1–Y5 planned utilisation trajectory.

Y1–Y3 sit below break-even (as expected for an early-stage TaaS build-out); Y4 is above; Y5 is comfortably above. The gap between the 122-day requirement and the Y1 plan of 40 days is the explicit financial quantification of the "TaaS burden" period discussed in Section 2.4.

3.2 NPV and IRR – the capital-budgeting view

Because the five-year operational FCFs are loss-making until Y4, accounting profit is an insufficient test of investability. The appropriate test is discounted cash flow [16]: can the project, including its terminal value, return more than the investor's hurdle rate?

Method. Free cash flows (pre-equity, post-CAPEX) are taken from Table 2.6. A terminal value is added to Y5 at the peer-group median EV/Revenue multiple of $2.5\times$ Y5 revenue [12, 17], yielding £5.2M. The project NPV at a discount rate r is then

$$\mathrm{NPV}(r) \;=\; \sum_{t=1}^{5} \frac{\mathrm{FCF}_t}{(1+r)^t} \;+\; \frac{\mathrm{TV}}{(1+r)^5}. \tag{3.2.1}$$

The project IRR is the discount rate at which NPV$= 0$.

Result. NPV at benchmark discount rates is tabulated in Table 3.1 and visualised in Figure 3.2. At the 20% VC-central rate the project NPV is £1.32M; at the 25% high-hurdle rate it is £0.97M; IRR is 55.2%.

Table 3.1: Project NPV and IRR, discounting the five-year free-cash-flow series plus a Y5 terminal value of £5.2M (peer-group EV/Revenue median of $2.5\times$ Y5 revenue). Free cash flows are taken from Table 2.6 (pre-equity, post-capex). The IRR is the discount rate at which the project's NPV equals zero.

Discount rate	Project NPV (£k)
10%	2,396
15%	1,785
20%	1,322
25%	968
30%	694
Project IRR	**55.2%**

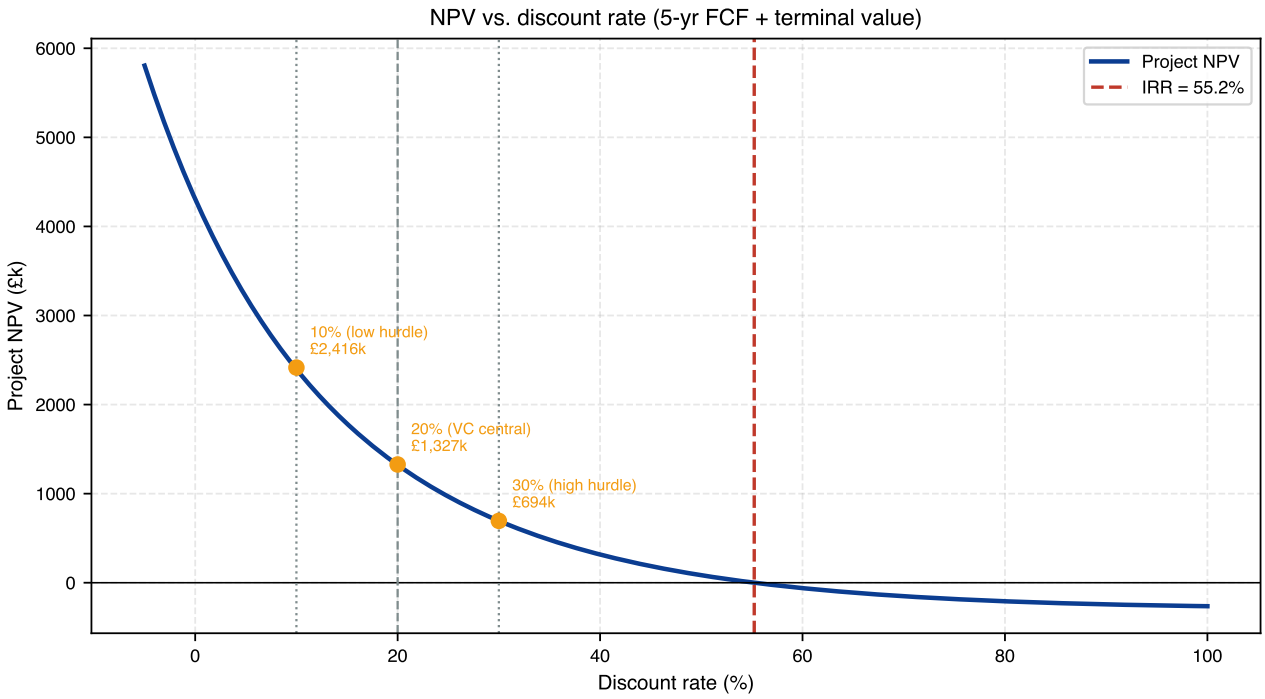

Figure 3.2: Project NPV as a function of discount rate. The 10% / 20% / 30% hurdle points are annotated; the IRR (55.2%) is marked with the red vertical dashed line. NPV remains positive at discount rates well above typical VC hurdles, driven by the present value of the Y5 terminal value.

Interpretation. The project clears standard VC-hurdle tests comfortably. Two caveats apply. First, **almost all of the NPV is present-value terminal value**, not operational cash flow: at 20% the PV of the £5.2M terminal alone is £2.1 M, against £0.8 M of negative PV from the Y1–Y3 cash outflows. NPV is therefore only as robust as the exit-multiple assumption. Second, the 55.2% IRR reflects that a TaaS project that turns cash-positive in Y5 is, arithmetically, a very-high-return asset if it can be sold at the terminal multiple; it does not imply year-by-year operating returns at that level.

3.3 Sensitivity analysis

Following the one-at-a-time methodology of the engineering chapter's sensitivity ranking [4], each driver is perturbed by $\pm 20\%$ ($\pm 50\%$ for inflation) and the resulting Y3 EBIT recorded.

Table 3.2: One-at-a-time sensitivity: impact on Year-3 EBIT (£k) of perturbing each driver. Drivers ranked by absolute swing; negative EBIT values shown in red. Mirrors the engineering sensitivity ranking in Harry Emes' engineering chapter.

Driver	Perturbation	Low	Base	High	Swing
Blended day rate (Y3)	±20%	-343	-156	31	**374**
Utilisation days (Y3)	±20%	-290	-156	-22	**268**
Fixed opex (all years)	±20%	-42	-156	-270	**-228**
Variable cost per day (Y3)	±20%	-103	-156	-209	**-106**
Capex per vehicle	±20%	-131	-156	-181	**-50**
Inflation rate	±50%	-136	-156	-176	**-40**

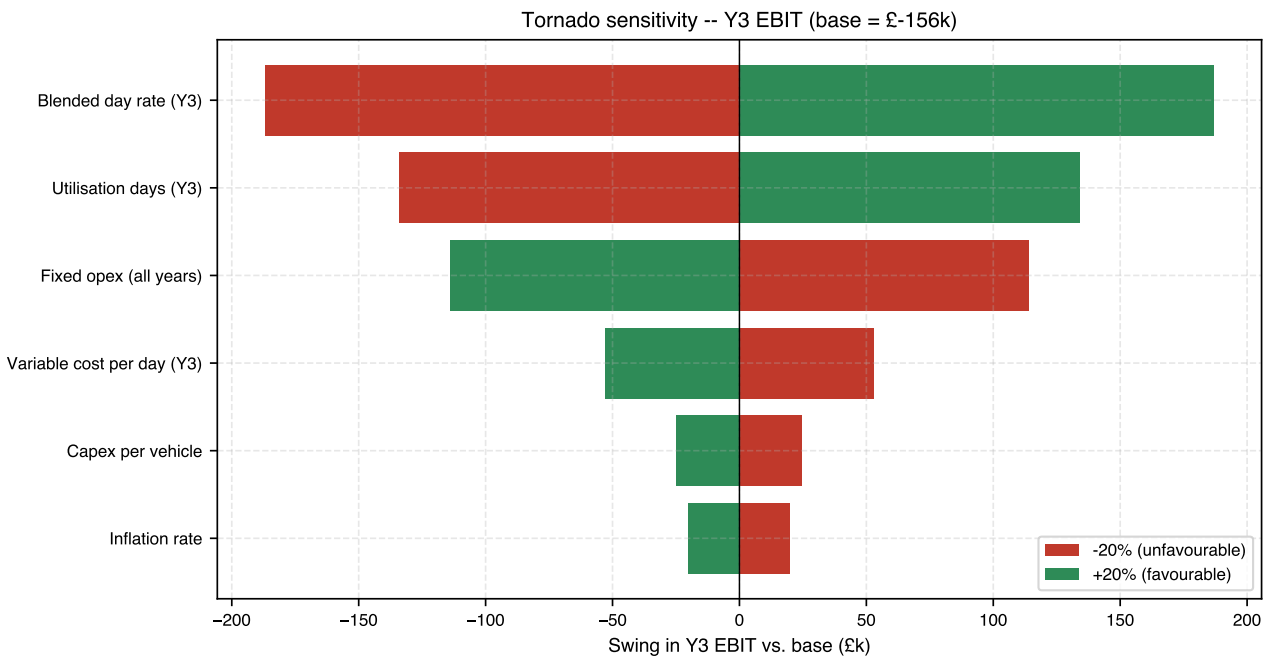

Figure 3.3: Tornado sensitivity of Y3 EBIT to $\pm 20\%$ (or $\pm 50\%$) perturbation of each driver. Blended day-rate and utilisation dominate; fixed opex is second-order; CAPEX and inflation are third-order.

Two implications follow. First, **pricing discipline matters more than cost discipline**: a 20% day-rate swing is worth £374 k of Y3 EBIT whereas a 20% fixed-opex swing is worth £228 k. Second, the two top drivers (day-rate and utilisation) are the two most directly in the operating team's control – management attention earns the highest marginal return there, rather than on cost-out or CAPEX engineering.

3.4 Scenario analysis

One-at-a-time sensitivity isolates drivers; scenarios combine plausible joint perturbations. Three are modelled:

- **Base**: the central plan.
- **Bull**: utilisation $+25\%$, day-rate $+10\%$, variable cost -5%. Represents strong product-market fit, two retainer conversions by Y3, and competitive pricing power.
- **Bear**: utilisation -30%, day-rate -10%, variable cost $+10\%$. Represents a slow AV investment cycle, discounted Bronze-heavy mix, and logistics / tyre cost inflation.

Table 3.3: Scenario analysis – net profit after tax (£k) in each of the three scenarios.

Scenario	Y1	Y2	Y3	Y4	Y5
Base	-402	-292	-156	12	293
Bull (+25% days, +10% rate, -5% var cost)	-307	-117	115	363	844
Bear (-30% days, -10% rate, +10% var cost)	-492	-462	-441	-400	-285

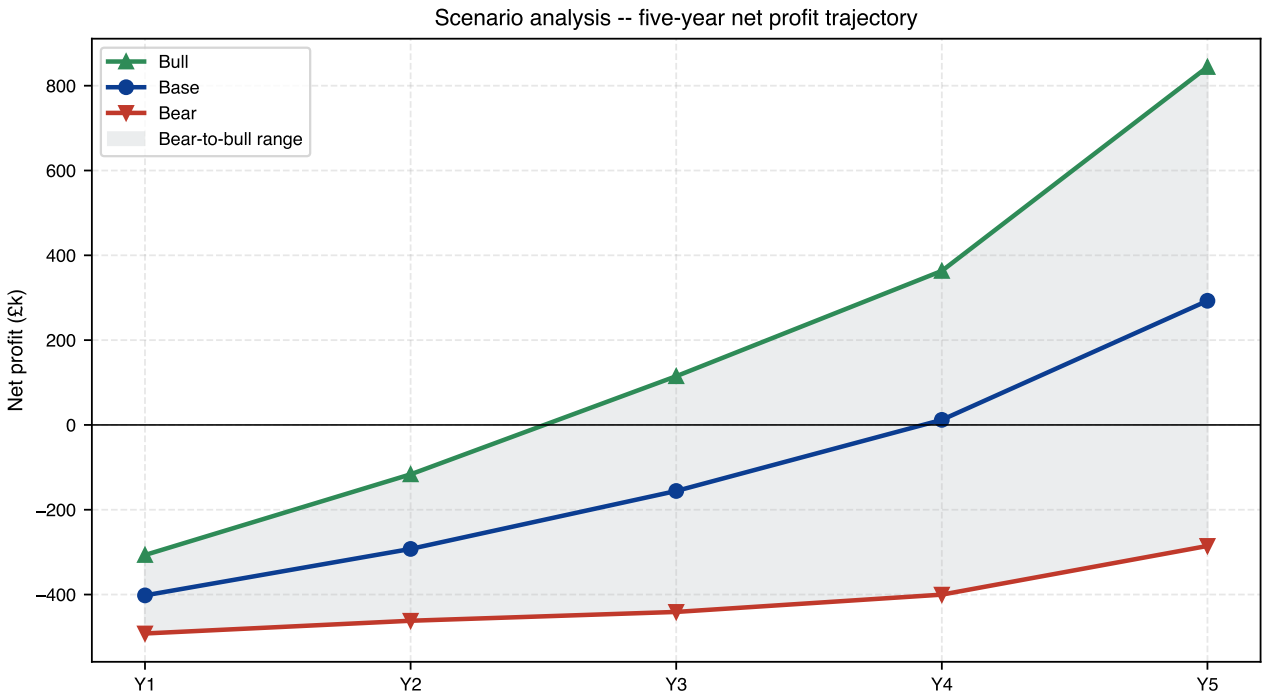

Figure 3.4: Scenario net-profit trajectories. The grey band is the bear-to-bull envelope. The Bull case reaches positive net profit in Y3 and ∼£700 k in Y5; the Bear case remains loss-making through Y5 and would necessitate a cost-base reset or a bridge round.

Chapter 4: Financial risk: the downtime case

McConnell's risk taxonomy [2] treats the industry-level risk environment. This chapter adds the financial-risk view: how robust are the NPV and EBIT forecasts to an *execution*-side failure on the fleet-maintenance dependency? In a TaaS business where revenue is utilisation-tethered (Section 1.1), this is the single most important execution-risk question.

4.1 Downtime sensitivity – quantifying the maintenance dependency

David's maintenance protocols target a $<$30-minute fleet-recovery time on any single failure, which underpins the planned utilisation arrays in Table 2.4. Downtime is modelled as a fractional reduction in billable days: a downtime rate of $d\%$ delivers $(1-d)\cdot D_t$ billable days in year t. The fixed cost base is unchanged (the crew is still paid while the vehicle is grounded), so every percentage point of downtime translates one-for-one into lost gross profit.

Table 4.1: Downtime sensitivity: effect of a persistent shortfall in billable days on revenue, EBIT and project NPV. A *downtime* of $d\%$ models the case where David's maintenance protocols deliver $(1-d)$ of the planned test days each year. Red values indicate an EBIT or NPV that has turned negative.

Downtime	Y1 days	Y5 days	Y5 rev.	Y3 EBIT	Y5 EBIT	NPV @ 20%
0%	40	220	2,090	-156	361	1,322
5%	38	209	1,986	-192	284	1,122
10%	36	198	1,881	-223	207	932
15%	34	187	1,776	-253	130	742
20%	32	176	1,672	-290	53	541
25%	30	165	1,568	-327	-24	338
30%	28	154	1,463	-357	-101	142

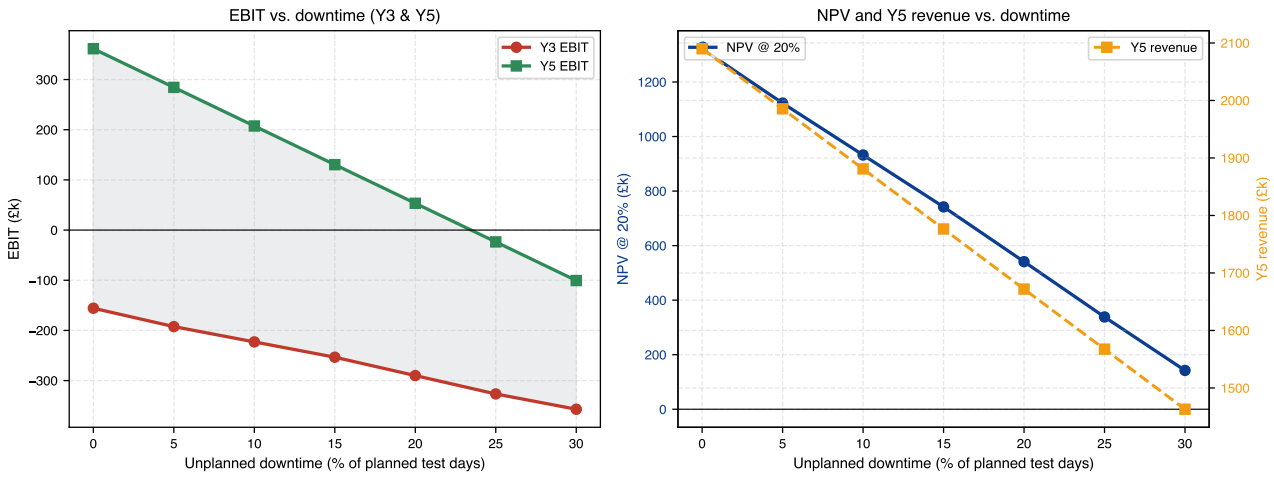

Figure 4.1: Downtime sensitivity. Left: Y3 (red) and Y5 (green) EBIT as a function of unplanned downtime. Y5 EBIT turns negative at ∼20% downtime. Right: project NPV at 20% discount rate (blue, left axis) and Y5 revenue (orange, right axis) vs. downtime. NPV stays positive through $> 30\%$ downtime, driven almost entirely by the preserved terminal-value contribution even at reduced Y5 revenue.

Interpretation.

- **EBIT is brittle.** Y3 EBIT deteriorates from −£156 k to −£357 k as downtime rises from 0 to 30%; Y5 EBIT turns negative at ∼20% downtime.
- **NPV is less brittle – but only as long as the exit multiple holds.** NPV@20% falls from £1.32 M to £0.14 M at 30% downtime, still positive but with little margin. If the downtime event also compresses the exit multiple (acquirers discount unreliability), NPV can flip negative at lower downtime levels.
- **The financial cost of a maintenance-protocol failure is asymmetric.** A 10% systematic downtime costs £190 k of Y5 revenue, £154 k of Y5 EBIT, and £390 k of NPV@20%. A 20% systematic downtime costs £418 k, £308 k, and £781 k respectively.

Implication for the maintenance budget. The financial cost of a 10% systematic downtime (£390 k NPV reduction) is more than double the total Y1 vehicle-maintenance contract (£20 k) and an order of magnitude above the incremental cost of a spare-vehicle strategy. Investment in maintenance capability should be benchmarked against this NPV-protection value, not the headline opex line.

4.2 Financial risk: cash-headroom assessment

The cash forecast (Section 2.6) identified end-Y2 cash balance of £90 k as the tightest point of the plan. A compounded shock at that moment – for example, a Series A close three months late plus

a reportable safety incident – is sufficient to push the business into bridge-round territory. Three specific financial exposures follow:

- **Series A timing.** Only £90 k buffer at Y2/Y3 boundary. Series A prep therefore starts at Month 15 to target a Q4-Y2 close; bridge-round authorisation is pre-negotiated with seed investors in the seed term sheet.
- **Reportable safety incident.** A single incident can increase insurance premium by 2–$5\times$ (shifting the insurance line from ∼£20 k/year to ∼£50–100 k/year) and close the OEM supplier-approval pathway – gating the Phase 3 Y4+ revenue ramp.
- **Anchor-customer concentration.** In Y1, $>$30% of revenue is from a single pilot customer. The Series A trigger therefore requires *two* retainer conversions, not one, to decouple the capital raise from single-customer concentration.

These financial-risk mitigations tie back to the execution plan but are deliberately framed here in cash terms, because cash-runway protection is the financial-evaluation lens that McConnell's industry-risk taxonomy does not cover.

Chapter 5: Funding strategy, valuation and financial KPIs

5.1 Funding strategy and use of funds

The plan requires a total of £2.5M of equity, in two rounds.

- **Seed, £1.0M at incorporation.** Used for Y1 vehicle build (£310k), Y1 workshop / data-platform setup (£50 k), and ∼18 months of runway (∼£600 k). Indicative pre-money valuation £4 M ⇒ 20% equity round.
- **Series A, £1.5M at Q4-Y2 / Q1-Y3.** Used for the second-vehicle build, Y3–Y4 headcount additions (BD/Sales, second test engineer), and a further ∼18 months of runway. Indicative pre-money valuation £6 M ⇒ 20% equity round.

The **Series A trigger is revenue-based, not runway-based**: $\geq$£400 k ARR and $\geq$ 2 retainer customers. Missing the trigger implies a bridge round from seed investors rather than a Series A at a compressed valuation; the seed term sheet should pre-authorise this bridge.

Unit economics. Using an average customer of ∼8 test days/year at a blended day-rate of £7.5 k, three-year average relationship, and 70% revenue-weighted gross margin:

$$\mathrm{LTV} = (8 \times 7{,}500) \times 3 \times 0.70 = £126k, \qquad \frac{\mathrm{LTV}}{\mathrm{CAC}} = 10.5\times, \qquad \mathrm{Payback} = 3.4 \text{ months.} \tag{5.1.1}$$

Both metrics clear standard B2B-services investability thresholds [18] by a wide margin; a doubling of CAC still clears the 3× LTV/CAC rule.

5.2 Valuation (comparable multiples)

Indicative Y5 valuation uses listed-peer EV/Revenue multiples, benchmarked against AB Dynamics (LSE: ABDP), Spirent Communications (LSE: SPT), and AVL List [12, 15].

Table 5.1: Comparable-multiples valuation. Year-5 forecast revenue (£2.1M) multiplied by peer EV/Revenue multiples gives the indicative Year-5 enterprise-value range.

Peer	EV / Revenue	Implied EV (£M)
AB Dynamics (LSE: ABDP, 2024)	3.5×	7.3
Spirent Communications (LSE: SPT)	2.2×	4.6
AVL List (private, Refinitiv est.)	1.8×	3.8
Peer-group median	2.5×	5.2

At the peer-group median of $2.5\times$ Y5 revenue, the indicative Y5 enterprise value is £5.2M; the AB Dynamics multiple (3.5×) yields £7.3 M. This range bracketed the terminal-value assumption used in the NPV analysis (Section 3.2); the NPV conclusion is robust within this band but sensitive to a compression below the peer-group floor (AVL at 1.8×).

5.3 Financial KPIs

Eight financial KPIs are tracked monthly (Table 5.2). The first six are operational and map directly to drivers in the financial model; the last two – *incident-free test days* and *unplanned fleet downtime* – are the execution-level signals that protect the financial plan's downside case described in Chapter 4.

Table 5.2: Financial KPI dashboard. Red-amber-green thresholds are set against the base-case financial plan; a red reading on any one triggers a board-level review.

KPI	Definition	Source
Revenue (monthly)	Invoiced revenue, accrual basis	Accounting system
Blended day-rate	Revenue / test days sold	Accounting
Test-days sold	Billable fleet days in month	CRM / ops
Gross margin	(Revenue − variable cost) / revenue	Accounting
Fixed-opex burn	Monthly fixed-opex spend	Accounting
Cash runway	Cash balance / monthly burn	Finance
Incident-free test days	Consecutive test days without reportable incident	Safety log
Unplanned downtime	Lost test days / planned test days	Maintenance log

Chapter 6: Conclusions and Recommendations

The AEB Rabbit TaaS business clears standard capital-budgeting thresholds in the base case: project NPV of £1.32M at a 20% VC-central discount rate, IRR of 55.2%, and break-even in Y4 at 122 test days/year. The financial structure is typical of a capex-inverted services build-out – loss through Y3, cash-thin at the Y2/Y3 boundary, and NPV dominated by the present value of the Y5 exit. Six recommendations follow, each paired with the Y1 financial action it drives.

1. **Raise £1.0M at incorporation and £1.5M in Q4-Y2 / Q1-Y3.** Series A trigger is revenue-based (>£400 k ARR, two retainer customers), not runway-based. *Y1 action*: close the seed round on the term sheet; pre-authorise seed bridge.
2. **Invest ahead of the maintenance-protocol budget.** A 10% systematic downtime costs £390 k of NPV – an order of magnitude above the baseline maintenance-opex line. *Y1 action*: scope a spare-vehicle strategy and benchmark it against the NPV-protection value, not the accounting opex.
3. **Prioritise pricing discipline over cost discipline in management incentives.** The sensitivity ranking identifies blended day-rate as the highest-leverage financial driver; the commercial team should be compensated on mix-shift to Silver/Gold, not only on volume. *Y1 action*: incentive plan pays on blended day-rate, not bookings.
4. **Gate Y4+ revenue on CAM Testbed UK / CCAV / Thatcham acceptance.** The terminal-value assumption (2.5× Y5 revenue) only holds if the platform is regulator-approved. *Y1 action*: commission the ISO / IEC 17025 gap-analysis and initiate the CAM Testbed UK relation-

ship.

5. **Protect the Y2/Y3 cash boundary at all costs.** Only £90 k of buffer; compound shocks can cascade into a bridge round at a compressed valuation. *Y1 action*: target Series A close at Q4-Y2, not Y3.
6. **Design the exit from Day 1.** Almost all of the NPV is terminal-value-driven; the operating company is therefore only as investable as the clarity of its trade-sale pathway to AB Dynamics / AVL / Spirent. *Y1 action*: include exit-readiness KPIs (ISO 17025 accreditation track, data-corpus size, OEM-framework pipeline) in the board pack from Month 1.

The TaaS financial structure is viable and defensible. The single biggest threat to financial returns is not the external market, the competitors, or the cost base – it is the execution of the maintenance protocol the model assumes. Protecting that assumption is both the cheapest and highest-leverage financial action the operator can take.

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